

# COBOL — Reference

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**COBOL** here is a free-format subset compiled with `lex` + `yacc` + `cc` (`cobol.l` / `cobol.y` lower COBOL to C; `cc` lowers the C to .NET IL). The four divisions map cleanly: `PROGRAM-ID` names it, `WORKING-STORAGE` builds the data, `PROCEDURE DIVISION` becomes the code. Each **paragraph** is emitted as a `public static` method on `CProgram`, so C#/VB.NET can invoke compiled COBOL.

Free-format (vs the columnar fixed format) is the escape from COBOL's column pathology: columns don't matter and `*>` starts a comment. Two COBOL-specific scanner problems are handled without lexer start-states: the **PICTURE clause** is grabbed as a single token (an internal `.` is kept only when glued to a following picture char, so the sentence-ending period survives), and a lone `.` is the statement **terminator**.

```
cobol prog.cob           # compile to prog.exe (native .NET executable)
cobol prog.cob -o app.exe # choose the output name
cobol lib.cob --dll       # compile to a library for C#/VB.NET to reference
```

## Quick Reference

```
IDENTIFICATION DIVISION.  PROGRAM-ID. NAME.
DATA DIVISION.            WORKING-STORAGE SECTION.
01 COUNTER  PIC 9(4) VALUE 0.          *> numeric (4 digits)
01 NAME     PIC X(20) VALUE "Ada".     *> text
01 AMOUNT   PIC 9(5)V99.               *> implied decimal point
01 PRETTY   PIC ZZ,ZZ9.99.            *> edited (zero-suppress, comma, point)
01 TABLE-X PIC 9(3) OCCURS 10 INDEXED BY IX.  *> array, subscript TABLE-X(i), 1-based
01 N        PIC 9(4) USAGE COMP.       *> usage (COMP/COMP-3/BINARY/DISPLAY)
01 AGE      PIC 9(3) VALUE 30.
      88 IS-ADULT VALUE 18 THRU 120.    *> condition-name (a boolean test)
01 REC.                      *> group item (record)
      05 FLD-A PIC X(8).
      05 FLD-B PIC 9(3).
PROCEDURE DIVISION.
      DISPLAY "text" FIELD.      MOVE x TO y.      ACCEPT y.      INITIALIZE y.
      MOVE ALL "*" TO y.      MOVE HIGH-VALUES TO y.      MOVE SPACES TO y.      MOVE y(2:3) TO z.
      MOVE REC1 TO REC2.      MOVE CORRESPONDING G1 TO G2.      FLD-A OF REC  *> group / OF
      ADD a b TO c [GIVING d] [ROUNDED]. SUBTRACT a FROM b. MULTIPLY a BY b. DIVIDE a INTO b.
      COMPUTE d ROUNDED = (a + b) * c.
      IF x IS NUMERIC / IS POSITIVE / IS NEGATIVE ...      IF cond THEN ... ELSE ... END-IF.
      EVALUATE n WHEN 1 ... WHEN 2 ... WHEN OTHER ... END-EVALUATE.
      PERFORM PARA.      PERFORM A THRU C.      PERFORM PARA 5 TIMES.      PERFORM PARA UNTIL cond.
      PERFORM VARYING i FROM 1 BY 1 UNTIL i > 10 ... END-PERFORM.
      SET cond-name TO TRUE.      SET i UP BY 1.      GO TO PARA.      STOP RUN.
      STRING a DELIMITED BY SPACE b DELIMITED BY SIZE INTO d END-STRING.
      UNSTRING s DELIMITED BY "," INTO p1 p2 END-UNSTRING.
      INSPECT s TALLYING n FOR ALL "x".      INSPECT s REPLACING ALL "x" BY "y".
      CALL "SUBPROG" USING a b c.      GOBACK.      *> subprogram with LINKAGE / USING
      FUNCTION UPPER-CASE(s) LOWER-CASE REVERSE LENGTH NUMVAL MOD MAX MIN INTEGER
```

## Data types

Every item has a **PICTURE** that gives its category and size:

- **9** numeric digit — **PIC 9(4)** is a 4-digit integer; **PIC 9(5)V99** has an implied decimal point (a fixed-point number).
- **X** any character, **A** letter — **PIC X(20)** is a 20-character string.
- **edited** (**Z** zero-suppress, **,** **.** insertion, **\$**, sign) — a display field a number is formatted into on **MOVE**.
- **S** marks a signed number; **V** an implied decimal point.

`OCCURS n` makes an item an array (subscripted `T(i)`, 1-based). `VALUE` gives an initial value. An **88-level** under an item is a *condition-name* — a named boolean.

## Statements / Commands

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`DISPLAY`, `ACCEPT`, `MOVE`, `INITIALIZE`, `SET`, the arithmetic verbs (`ADD`, `SUBTRACT`, `MULTIPLY`, `DIVIDE`, `COMPUTE`), `IF/ELSE/END-IF`, `EVALUATE`, `PERFORM` (out-of-line, `THRU`, `N TIMES`, `UNTIL`, `VARYING`, and inline `... END-PERFORM`), `GO TO`, `CALL ... USING`, `GOBACK` / `EXIT PROGRAM`, the string verbs `STRING` / `UNSTRING` / `INSPECT`, and `STOP RUN`. A sentence is one or more statements ended by a period; block statements use scope terminators (`END-IF`, `END-PERFORM`, `END-EVALUATE`).

## Functions

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COBOL has two units of code. A **paragraph** is a name followed by sentences; `PERFORM` runs it and returns (`PERFORM A THRU C` runs a range). A **subprogram** is a separate program with a `LINKAGE SECTION` and `PROCEDURE DIVISION USING ...`, invoked with `CALL "NAME" USING args` and returning via `GOBACK` / `EXIT PROGRAM`; its arguments are passed **by reference**, so the caller sees updates. Multiple programs can live in one file, each ended by `END PROGRAM name`. Each paragraph compiles to a `public static void` method on `CProgram`, which is what makes a compiled COBOL program callable from C#/VB.NET (Activity 9). Built-in intrinsics: `FUNCTION UPPER-CASE` / `LOWER-CASE` / `REVERSE` / `LENGTH` / `NUMVAL` / `INTEGER` / `MOD` / `MAX` / `MIN`.

## Input / Output

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`DISPLAY a b c` writes the operands (no separators) and a newline; numeric items print per their PICTURE (zero-padded), edited items print their formatted text. `ACCEPT x` reads a line/number into `x`.

**File I/O** (line-sequential). A file is declared in the ENVIRONMENT and DATA divisions and driven by `OPEN` / `READ` / `WRITE` / `CLOSE`:

```

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT EMP-FILE ASSIGN TO "employees.dat"
        ORGANIZATION IS LINE SEQUENTIAL
        FILE STATUS IS WS-ST.
DATA DIVISION.
FILE SECTION.
FD EMP-FILE.
01 EMP-REC.
    05 EMP-ID    PIC 9(4).
    05 EMP-NAME  PIC X(12).
    05 EMP-SAL   PIC 9(6).
...
    OPEN OUTPUT EMP-FILE.    WRITE EMP-REC.    CLOSE EMP-FILE.
    OPEN INPUT EMP-FILE.
    READ EMP-FILE.
    PERFORM UNTIL WS-ST = "10"
        ... process EMP-REC ...
        READ EMP-FILE
    END-PERFORM.
    CLOSE EMP-FILE.

```

Each record's elementary fields are packed/unpacked to fixed-width columns. `READ` sets the `FILE STATUS` field to `"00"` (read) or `"10"` (end-of-file), which the loop tests — the priming-read idiom above. `READ ... INTO ws-rec` also copies the record into a working-storage group. ( `READ` uses `FILE STATUS` rather than an inline `AT END` clause — see *Subset boundaries*.)

## Graphics

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None. COBOL's "picture" is the **PICTURE clause**: Activity 8 below uses edited pictures to format a money report — COBOL's native idea of drawing.

## Notes

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- **Free-format only.** `*>` comments; columns don't matter. A period ends a sentence.
- **Keywords are reserved** (a large COBOL reserved-word set).
- The `PICTURE` clause is captured as one token; `(n)` repeat-counts and an internal `.` are supported.
- Numeric items are 32-bit integers ( `9` , `S9` ); `V` -decimals and edited fields are computed in double precision for display.
- Paragraphs are emitted as functions and `main` calls them in source order, so a `STOP RUN` that halts before fall-through behaves like real COBOL.

## Subset boundaries

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A broad core. **Included** (beyond the basics): line-sequential file I/O

( **SELECT** / **ASSIGN** / **FD** / **OPEN** / **READ** / **WRITE** / **CLOSE** / **FILE STATUS** ); group items with a real record hierarchy and **group-level MOVE**; **reference modification** ( **x(start:len)** ) as source *and* target; **OF** / **IN** qualification of duplicate field names; **MOVE** / **ADD CORRESPONDING**; **USAGE** ( **COMP** / **COMP-3** / **BINARY** / **DISPLAY** , stored as the natural .NET type); **ROUNDED**; **OCCURS ... INDEXED BY**; **STRING** / **UNSTRING** / **INSPECT**; intrinsic **FUNCTION** s; and subprograms ( **LINKAGE** / **PROCEDURE DIVISION USING** / **CALL** ).

**Not included** (a managed-runtime/LALR-front-end takes precedence — your stated constraint): fixed (columnar) source format; indexed/relative file organizations and **REWRITE** (line-sequential only); the report writer; **COPY** copybooks; **REDEFINES** (overlapping storage doesn't fit the typed model); **SEARCH** and **EVALUATE TRUE** (their **cond** /statement-list-in-clause shape conflicts with our small LALR table); a **READ** inline **AT END** / **NOT AT END** clause (use **FILE STATUS** instead, as shown above). **INSPECT ... LEADING** is treated like **ALL**. C#/VB.NET interop drives paragraph entry points; the **CALL ... USING** mechanism (by-reference parameters) is for COBOL-to-COBOL calls within the run unit.

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## Tutorial

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Every example was compiled and run with **cobol**; the output shown is real.

### 1. Your first program

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. HELLO.  
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 WS-NAME PIC X(5) VALUE "World".  
01 WS-COUNT PIC 9(3) VALUE 0.  
PROCEDURE DIVISION.  
    DISPLAY "Hello, " WS-NAME.  
    MOVE 5 TO WS-COUNT.  
    ADD 10 TO WS-COUNT.  
    DISPLAY "Count is " WS-COUNT.  
    STOP RUN.
```

```
Hello, World  
Count is 015
```

Note **015** — a **PIC 9(3)** item always displays its full three digits, zero-padded. That PICTURE-driven formatting is COBOL's signature.

## 2. Variables and data types

Data lives in WORKING-STORAGE, each item introduced by a **level number** and a **PICTURE**:

```
01 CUST-NAME PIC X(20).      *> 20-character text
01 BALANCE   PIC 9(6)V99.    *> number with two implied decimals
01 COUNTER   PIC 9(4) VALUE 0.
```

**9** is a digit, **X** a character, **V** an implied decimal point, **VALUE** an initial value. The PICTURE fixes both the *type* and the *width*.

## 3. Flow control

Block **IF**, multi-way **EVALUATE**, and the **PERFORM** loops:

```
EVALUATE GRADE
  WHEN 1 DISPLAY "grade one"
  WHEN 2 DISPLAY "grade two"
  WHEN OTHER DISPLAY "other grade"
END-EVALUATE.
IF TOTAL > 10 THEN
  DISPLAY "big"
ELSE
  DISPLAY "small"
END-IF.
```

With **GRADE = 2** and **TOTAL = 15**:

```
grade two
big
```

**EVALUATE** is COBOL's switch; block statements close with scope terminators ( **END-IF** / **END-EVALUATE** ) rather than a period.

## 4. Arrays

**OCCURS** makes a table; subscripts are 1-based and written with parentheses:

```

01 NUMS  PIC 9(2) OCCURS 5.
01 I      PIC 9(2) VALUE 0.
01 TOTAL PIC 9(4) VALUE 0.
...
PERFORM VARYING I FROM 1 BY 1 UNTIL I > 5
    MOVE I TO NUMS(I)
END-PERFORM.
PERFORM VARYING I FROM 1 BY 1 UNTIL I > 5
    ADD NUMS(I) TO TOTAL
END-PERFORM.
DISPLAY "Sum of 1..5 is " TOTAL.

```

```
Sum of 1..5 is 0015
```

`NUMS(I)` indexes the table; the inline `PERFORM VARYING` is COBOL's counted loop.

## 5. Subroutines and functions — paragraphs and PERFORM

COBOL structures code into **paragraphs**, invoked with `PERFORM`:

```

PROCEDURE DIVISION.
MAIN-PARA.
    PERFORM ADD-PARA VARYING I FROM 1 BY 1 UNTIL I > 5.
    DISPLAY "Total is " TOTAL.
    STOP RUN.
ADD-PARA.
    ADD I TO TOTAL.

```

```
Total is 0015
```

`PERFORM ADD-PARA VARYING ...` runs `ADD-PARA` once per value of `I`. `MAIN-PARA` ends in `STOP RUN`, so control never falls into `ADD-PARA` on its own — exactly COBOL's fall-through-unless-halted model.

`PERFORM A THRU C` runs a *range* of paragraphs.

For reusable code with parameters, COBOL uses a **subprogram** — a separate program with a `LINKAGE SECTION` and `PROCEDURE DIVISION USING`, called with `CALL` (arguments by reference) and returning with `GOBACK`. Multiple programs share one file, each ended by `END PROGRAM`:

```

PROGRAM-ID. MAINPROG.

...
    CALL "ADDER" USING A B RESULT.
    DISPLAY "Result is " RESULT.
    STOP RUN.
END PROGRAM MAINPROG.

IDENTIFICATION DIVISION.
PROGRAM-ID. ADDER.
DATA DIVISION.
LINKAGE SECTION.
01 X PIC 9(4).
01 Y PIC 9(4).
01 Z PIC 9(6).
PROCEDURE DIVISION USING X Y Z.
    COMPUTE Z = X + Y.
    GOBACK.
END PROGRAM ADDER.

```

With **A=7**, **B=5** :

```

Calling ADDER with 0007 and 0005
Result is 000012

```

**ADDER** writes its result back through **Z**, which is bound by reference to the caller's **RESULT** — the defining behaviour of a COBOL subprogram.

## 6. Memory management

Storage is **static** — there is nothing to allocate or free:

- Every WORKING-STORAGE item is a fixed-size field that exists for the whole run.
- **OCCURS n** reserves **n** fixed slots; **PIC X(n)** reserves **n** characters.
- **VALUE** sets the initial contents once at start-up.

No heap, no pointers ( **REDEFINES**, dynamic tables, and **LINKAGE** are subset boundaries).

## 7. Strings, condition-names, and editing

Text is **PIC X(n)**; **MOVE** copies with space-padding/truncation to the field width. An **88-level** gives a condition a name:



```
01 AGE PIC 9(3) VALUE 25.  
88 IS-ADULT VALUE 18 THRU 120.  
...  
IF IS-ADULT  
    DISPLAY "adult"  
END-IF.
```

```
adult
```

`IS-ADULT` reads as a boolean but is really the test `AGE ≥ 18 AND AGE ≤ 120` — COBOL's way of naming a condition. `SET IS-ADULT TO TRUE` assigns the underlying value.

COBOL's string verbs build and take apart fixed-width fields. `STRING` concatenates (with `DELIMITED BY SPACE` to trim padding), `UNSTRING` splits on a delimiter, `INSPECT` counts/replaces, reference modification `field(start:len)` takes a substring, and `MOVE ALL "*" fills:`

```
STRING FIRST-N DELIMITED BY SPACE " " DELIMITED BY SIZE LAST-N DELIMITED BY SPACE  
    INTO FULL-N END-STRING.  
DISPLAY "Full: [" FULL-N "]".  
UNSTRING CSV DELIMITED BY "," INTO PART1 PART2 END-UNSTRING.  
MOVE FULL-N(1:4) TO SUB.  
INSPECT TXT TALLYING CNT FOR ALL "S".  
INSPECT TXT REPLACING ALL "S" BY "*".  
DISPLAY "Upper: " FUNCTION UPPER-CASE("hello world").
```

```
Full: [JOHN SMITH]  
Part1: [alpha    ] Part2: [beta      ]  
Upper: HELLO WORLD
```

(with `CSV = "alpha,beta"`, `TXT = "MISSISSIPPI"` → `CNT` becomes 4 and `TXT` becomes `MI**I**IPPI`.)

## 8. Drawing a picture — a formatted report

COBOL's "picture" is the **PICTURE clause**; its drawing is a *formatted report*. An edited field (`ZZ,ZZ9.99`) turns a raw number into aligned, comma-grouped money:

```

01 A      PIC 9(6)V99 VALUE 1234.50.
01 B      PIC 9(6)V99 VALUE 99.99.
01 C      PIC 9(6)V99 VALUE 12000.00.
01 GRAND  PIC 9(7)V99 VALUE 0.
01 P      PIC ZZ,ZZ9.99.
PROCEDURE DIVISION.
    MOVE A TO P.
    DISPLAY "Item 1: $" P.
    MOVE B TO P.
    DISPLAY "Item 2: $" P.
    MOVE C TO P.
    DISPLAY "Item 3: $" P.
    COMPUTE GRAND = A + B + C.
    MOVE GRAND TO P.
    DISPLAY "Total : $" P.
    STOP RUN.

```

```

Item 1: $ 1,234.50
Item 2: $    99.99
Item 3: $12,000.00
Total : $13,334.49

```

The **Z**s suppress leading zeros to spaces, the comma appears only once digits start, and the decimals line up — the report "draws" itself through the PICTURE.

## 9. Calling COBOL from C# / VB.NET

Compile a COBOL file as a library; each paragraph becomes a **public static** method on **CProgram** :

```

IDENTIFICATION DIVISION.
PROGRAM-ID. COBLIB.
PROCEDURE DIVISION.
SHOW-BANNER.
    DISPLAY "≡≡≡ Daily Report (compiled from COBOL) ≡≡≡".
SHOW-FOOTER.
    DISPLAY "--- end of report ---".

```

```
cobol lib.cob --dll          # → coblib.dll
```

A C# program that references **coblib.dll** (and **CRuntime.dll** ) calls the paragraphs directly:

```

CProgram.pg_SHOW_BANNER();
Console.WriteLine(" ... (C# does its own work here) ... ");
CProgram.pg_SHOW_FOOTER();

```

```
≡≡≡ Daily Report (compiled from COBOL) ≡≡≡  
... (C# does its own work here) ...  
--- end of report ---
```

The COBOL runs as ordinary .NET methods interleaved with C#. Passing arguments in and out (a **LINKAGE SECTION** with **PROCEDURE DIVISION USING**) is the natural next step (*Subset boundaries*), along with file I/O and **STRING** / **UNSTRING** for fuller COBOL.